Plan — R11.4A: Target-Conditioned Delivery + Settle Teacher (with phase energy instrumentation)

Replace the frozen canonical R2 downstream with a per-instance solver that re-derives the post-capture delivery/target-entry/settle trajectory for the *actual* coin/target geometry. Measure delivery+settle energy before optimizing it. Generation/diagnosis only — no BC/RL/refinement. Pilot, then **HALT**.

HyMeKo coin-delivery campaign (`feature/r11-4a-target-conditioned-delivery-teacher`, from R11.3 `5b3995`

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Abstract

R11.3 proved the demonstration *infrastructure* (exact-zero certificates, deployed RRT reach, valid precontact handoffs, real target relocation) and localized the blocker: the **frozen canonical downstream (R2) does not generalize** — 8/64 strict-K6, failures concentrated in *delivery* (21) and *target-entry/settle* (33 scenario-best; 68 at the all-attempt level) with **0 `precontact_handoff_invalid`**. A capture-phase proxy is *suggestive* (successes have lower terminal velocity scale) but **unproven** without delivery/settle energy. R11.4A replaces the frozen R2 delivery with a **target-conditioned delivery+settle teacher**: a per-scenario CEM over the structured `forward_displacement.rollout_primitive` (deliver mode), which already pushes/brakes/releases/settles the coin toward `direction_to_zone` (the relocatable target) and monitors the frozen K6 certificate. R11.4A also extends `MeasuredEnergyLedger` across capture→delivery→target-entry→settle so the energy-surplus mechanism becomes a *measured* claim. **Measure before optimizing; promote PLAUSIBLE→SUPPORTED only under matched scenarios.** A bounded pilot (12 settle + 8 delivery + 2 control), then **HALT**. Verdict: `R11_4A_TARGET_CONDITIONED_DELIVERY_TEACHER_PILOT_PASS` or `R11_4A_PIPELINE_PASS_TEACHER_RECOVERY_INSUFFI`

1 Binding review refinements

(1) Prove target-conditioning, pre-pilot. Contract tests: relocating the target changes `direction_to_zone`; rotating it rotates the primary push; braking + the K6 monitor use the *same* relocated zone; no hidden canonical target / absolute R2 reference; the relocated target is the rollout's *terminal* goal, not a report field. Gates `TARGET_RELOCATION_PROPAGATES_THROUGH_DELIVERY_PASS`, `NO_CANONICAL_DELIVERY_ORBIT_DEPENDENCY_PASS`.
(2) Phase A opens dimensions incrementally: freeze the early push {squeeze, forward, balance}; the CEM searches *only* {brake onset (`ramp`), brake gain, release timing, entry-speed target, settle damping, settle dwell, rebound suppression}. Only if that is insufficient open {forward magnitude, delivery duration, direction correction, balance, ramp}. This distinguishes energy-shedding from "needed a new orbit." **(3) Energy is NOT an optimization objective in this pilot** — the CEM is driven by K6/safety/progress/overshoot (the existing `score`); the energy ledger is observer/diagnostic only, so a lower target-entry energy in the winners is *discovered*, not fitted. Energy becomes an explicit objective no earlier than R11.8. **(4) Instrumentation proven non-invasive:** the phase hook/energy read must not change control-call order, physics-step count, event timing, RNG consumption, or the trajectory. Gate `ENERGY_INSTRUMENTATION_NONINVASIVE_PASS` — same seed+params, hook vs. no-hook rollout bit-exact (or within a pre-declared tolerance). (The `rollout_primitive frame_hook` is contractually non-behavioural; this gate proves it.) **(5) Full provenance per attempt:** coin/target geometry, relative goal vector, capture outcome, delivery + settle params, phase-event timestamps, target-entry state, energy ledger, K6 result, failure class, teacher seed, provenance hash — BC needs the geometry/handoff/energy context,

not just the actions. **Settle-extension coordinate** (separate block, normalized + scenario-relative): `entry_speed_target`, `brake_onset_distance` (fraction of remaining delivery distance, not absolute mm), `settle_damping`, `settle_dwell`, `rebound_gain`. **Pilot selection**: the 12 settle-failures span C0–C3, short + long delivery distances, differing target directions, and low + high capture `vel_scale` — not one easy settle cured twelve times.

2 Scope, non-goals

Goal. A per-instance target-conditioned delivery+settle solver that re-derives the post-capture trajectory from the current coin pose/velocity, target pose, relative goal vector, robot + contact state, and handoff descriptor — never reusing a frozen canonical delivery orbit as the executable solution. **Non-goals (hard):** no BC, no SAC/TD3/PPO, no policy assimilation, no reach refinement, no Hamiltonian reward shaping. **Roles unchanged:** RRT deployed; every CEM/search call is a labelled training teacher in provenance; no teacher-free label. **Two-level energy contract preserved:** measured ledger + residuals only; `conservation_verdict()` continues to refuse (`HAMILTONIAN_CONSERVATION_PASS` is R11.8).

3 Established (R11.3) — do not overclaim

Exact-zero certificates work; deployed RRT reach generalizes over the admissible coin region; precontact handoffs valid; target relocation real + correctly represented (C3 `entry_dtz` 50–90 mm matches r=5–9 cm); failures localize to delivery + target-entry/settle; the frozen downstream does not generalize. The capture-phase proxy (successes `vel_scale` 1.56 vs settle-failures 1.80; reach work + capture direction flat) **supports but does not prove**: *excess/poorly-directed energy at target entry contributes to delivery/settle failure.* **Not a confirmed mechanism until delivery+settle energy is measured directly.**

4 Teacher pipeline

EXACT_ZERO_HOME → deployed RRT reach → precontact handoff → capture solve →
target-conditioned delivery solve → target-entry braking → release / contact-mode →
***settle solve** → strict K6.*

The delivery+settle stages are re-derived per instance. **Grounding:** the executable primitive is the existing `forward_displacement.rollout_primitive` in `deliver` mode — its PUSH/BRAKE/RELEASE schedule already aims at `direction_to_zone` (relocatable target), brakes on measured coin speed, and monitors the frozen K6 certificate (`dtz`≤CENTER_TOL ∧ speed<SETTLE_VEL, HELD_DWELL, touched). R11.4A wraps it in a per-scenario CEM search and **replaces `FrozenDownstream.deliver` (R2)** with it. See `plan.tikz`.

5 Searchable teacher parameters

A structured coordinate (bounded, inside safety/torque/slew/collision/contact contracts), mapping onto the primitive + settle extension: **delivery direction** (target-conditioned push via `e_par`$=\hat{u}_{\text{zone}}$, lateral balance correction, goal-vector alignment); **magnitude/duration** (forward force/torque scale, transport duration / ramp, velocity target, acceleration profile); **target-entry braking** (braking onset by remaining distance, brake gain, desired entry speed, velocity-feedback term); **release/contact** (release timing, preload reduction, contact-mode continuation vs. switch, optional guided coast); **settle** (terminal-velocity target, settle damping, dwell duration, rebound suppression, strict-K6 dwell verification). No raw per-step torques.

6 Delivery + settle energy instrumentation

Extend `MeasuredEnergyLedger` across capture / delivery / target entry / settle (phase-marked frame hook on the rollout). Record per phase: robot E_k; coin translational E_k; coin rotational E_k (if available); W_+; W_-; contact impulse; impact-energy proxy; dissipation proxy; and energy *after capture*, *at target entry*, *at braking onset*, *at release*, *at settle start*, *at K6 / terminal failure*; numerical residual. Derived: $T_{\text{coin,entry}}$, H_{entry}, $\Delta H_{\text{capture}\to\text{entry}}$, $\Delta H_{\text{entry}\to\text{settle}}$, and target-directed-coin-energy / W_+. The certificate still asserts only `ENERGY_LEDGER_COMPLETE`/`ENERGY_BALANCE_RESIDUAL_RECORDED`; `conservation_verdict()` refuses.

7 Recovery sequence

Phase A — settle failures first (cheapest recoverable: coin already in the vicinity). Search braking onset, target-entry speed, release timing, damping, dwell, rebound suppression. Compare frozen canonical downstream vs. target-conditioned teacher; measure recovery-to-K6, $T_{\text{coin,entry}}$, overshoot, rebound, settle time, actuator work, safety. **Phase B — delivery failures** (after settle recovery verified): search push direction, transport duration, force/torque magnitude, mid-course correction, braking profile. **Phase C — capture failures** (deprioritized; kept a distinct class; not until delivery+settle coverage improves).

8 Matched mechanism analysis

No pooled energy comparisons. Compare the *same* coin/target scenario across teacher variants, within matched delivery-distance bands, matched target-direction bands, matched initial handoff quality. The energy-surplus hypothesis is promoted PLAUSIBLE→SUPPORTED *only if* recovery is associated — within matched scenarios — with reduced $T_{\text{coin,entry}}$ (or improved target-directed energy ratio), reduced overshoot/rebound, improved strict-K6, and no safety regression.

9 Pilot scope + gates

Scope: ∼12 scenario-best settle failures + 8 delivery failures + 2 success controls (canonical/translated). Per scenario: retain the frozen baseline; run $\leq$ 5 teacher seeds / search restarts; retain *every* attempt; never overwrite failures with later successes; link all to provenance hashes. Bounded search only. **Gates:** exact-zero certificate preserved; RRT deployed; all teacher calls labelled in provenance; delivery uses the actual coin+target geometry; target-entry + settle energy ledgers complete; no conservation claim; failure classes distinct; serialization + replay pass; no safety/boundary regression. **Scientific recovery gate:** $\geq$ 50% of selected settle-failure scenarios recover to strict K6. **Diagnostic delivery gate:** $\geq$ 1 previously-failed noncanonical delivery scenario recovers. **Verdict:** `R11_4A_TARGET_CONDITIONED_DELIVERY_TEACHER_PILOT_PASS`; if infra passes but recovery does not, `R11_4A_PIPELINE_PASS_TEACHER_RECOVERY_INSUFFICIENT` — stop + report honestly.

10 Full coverage target (after pilot PASS only)

Extend across the 64-scenario bank; require $\geq$ 70% admissible scenarios with $\geq$ 1 strict-K6; $\geq$ 50% positive coverage in each C0–C3 stage; positives across multiple coin+target regions; disjoint train/dev/test positives; no geometric region represented by a single canonical orbit. Verdict `TARGET_CONDITIONED_DELIVERY_TEACHER_C`
If it fails, do not start BC.

11 Affected files (new; from `5b3995e1`; no CORE.YAML items)

`hymeko_rl/coin_delivery/delivery_teacher/` (`solver.py` target-conditioned delivery+settle CEM over the `rollout_primitive` coordinate; `phase_energy.py` phase-marked energy ledger extension + derived quantities; `record.py` delivery-attempt record (extends the R11.3 schema with delivery/settle/energy fields + baseline vs re-solved); `matched_analysis.py` matched-scenario comparison; `failure_transition.py` baseline→re-solved failure-class transitions). `hymeko_rl/experiments/r11_4a_pilot.py`, `r11_4a_report.py`. `hymeko_rl/tests/test_r11_4a_delivery_teacher.py`. Reuse `forward_displacement.rollout_primitive` (the target-directed delivery/settle primitive — *do not reimplement*), `demo_bank` (scenario/record/store), `ir/ir_adapter`, `coin_zero_home_rrt`, `moving_precapture.plan_capture` (capture).

12 Reports, visualizations, validation

Reports: `reports/2026-07-30-r11-4a-target-conditioned-delivery-teacher.{md,json}` — recovery by original failure class, scenario-wise teacher success, stage coverage, target-entry energy distributions, matched SUCCESS vs SETTLE comparisons, braking/release parameter distributions, overshoot/rebound + settle-time stats, $W_{\pm}$, residual completeness, exact claims/non-claims. **Figures:** coin/target recovery map, failure-class transition matrix, target-entry E_k baseline vs re-solved, overshoot/rebound, settle time, W_{+}, energy traces for representative success/failure pairs. **Tests:** goal-vector propagation; relocated target honored throughout delivery; delivery energy-ledger boundaries; target-entry event detection; braking-onset / release / settle event logging; teacher provenance labelling; matched-scenario analysis; no conservation verdict; replay + serialization; failure-class stability. Run R11.2 + R11.3 + R11.4A tests + targeted coin-delivery regressions; `ruff`, `mypy --strict`, `radon`. Do not weaken existing tests.

13 Stop logic

Production-scale smoke (1 scenario) before any queue; reconcile the wall estimate ($> 2\times$ rule). Run the bounded pilot, verify the gates, write the pilot report, commit only then, push fast-forward, and **HALT** for review. **No R11.4B BC or R11.5 RL until the full-coverage gate passes.** *Measure delivery+settle energy before optimizing it.* **This plan is the R11.4A boundary-0 artifact — no solver code begins until it is approved.**

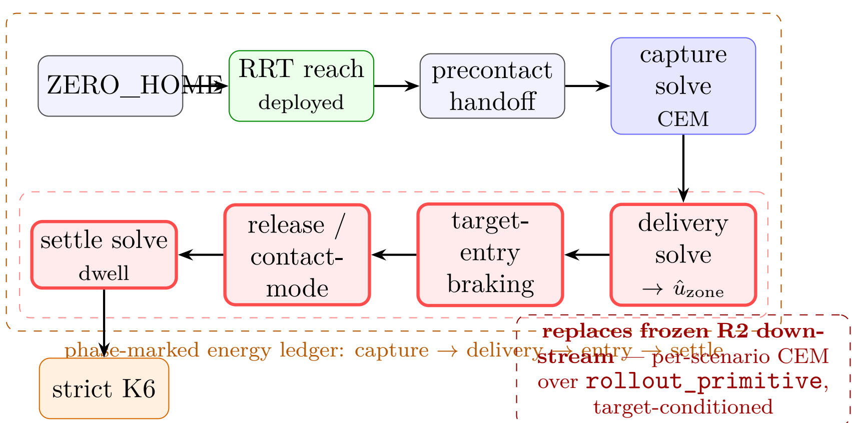

Figure 1: R11.4A pipeline. Green = deployed RRT reach; blue = existing capture teacher; **red = the new target-conditioned delivery+settle CEM** (over `rollout_primitive`, aimed at `direction_to_zone`) that replaces the frozen R2 downstream; orange = phase-marked energy ledger spanning capture→delivery→entry→settle.